Phonological Assembly Circuits in GPT-2: How Transformers Compute Sound Structure from Orthography

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Abstract

No published work has performed circuit-level causal analysis on phonological operations in transformer language models. We identify and validate circuits for six distinct phonological operations in GPT-2 Small, ranging from stored lexical lookup (rhyming hypocorism: Richard → Dick) to cross-boundary phoneme fusion (oronym: "Bill Ding" → "building"). Each operation requires a linguistically distinct computation — irregular lookup, regular truncation, grapheme-to-phoneme mapping, juncture reanalysis, homophone recognition, and reverse folk etymology — and we find that each activates a *distinct* circuit architecture with [PENDING: N–M] heads per circuit.

Using weight-space circuit discovery (Tower, 2026), EAP, and activation patching, we show that: (1) the universal backbone tier (L0H8, L0H9, L0H11) participates in all phonological circuits, consistent with the "operating system" hypothesis; (2) the oronym circuit — which must fuse phonemes across a word boundary — uses [PENDING: specific mid-layer heads in layers 3–8] that have not been identified in any prior circuit; (3) compound phonological puns ("Richard Tater" → "Dictator") activate the expected composition of component circuits, and ablating either component breaks the compound.

The oronym circuit is the central novel finding: it demonstrates that GPT-2 implements cross-boundary phonological assembly from purely orthographic input, a capability previously demonstrated only behaviorally (Liao and Shi, 2026; Suvarna et al., 2024) but never localized to specific attention heads.

1 Introduction

The mechanistic interpretability literature has characterized circuits for syntactic operations (IOI, SVA, reflexive anaphora) and simple retrieval tasks (induction, greater-than). But the space of *phonological* circuits — computations that operate on sound structure rather than syntactic structure — is entirely unexplored at the circuit level.

This gap matters because phonological computation in transformers is surprising: BPE tokenization encodes orthography, not phonology, yet language models demonstrably encode phonological structure in their hidden states (Liao and Shi, 2026). Linear probes on GPT-2's middle layers achieve 63–79% accuracy on rhyme detection, and larger models perform above chance on grapheme-to-phoneme conversion (Suvarna et al., 2024). The question we address is: *which specific attention heads compute this phonological knowledge, and do different phonological operations use different circuits?*

The phonetic name pun as a natural task. We use phonetic name puns as our task family because they decompose into well-specified linguistic operations with clear minimal-pair evaluations. "Richard Tater → Dictator" requires two operations: (Op 1) rhyming hypocorism (Richard → Dick) and (Op 4) oronym/juncture reanalysis (Dick Tater → Dictator). By testing each operation independently, we isolate the circuit for each computation.

Contributions.

1. Six linguistically specified circuit tasks with minimal-pair datasets ([PENDING: total N] pairs across 6 operations).
2. Circuit discovery for each operation using EAP, activation patching, and weight-space features.
3. The first identification of cross-boundary phoneme fusion heads in any transformer model (the oronym circuit).
4. Cross-reference with the GPT-2 acronym circuit (García-Carrasco et al., 2024): Op 3 (initialism) [PENDING: shares / does not share] heads with their letter-mover circuit.
5. Compositional validation: compound puns activate the expected combination of component circuits.

2 Operations and Datasets

2.1 Linguistic Classification

Each operation is defined by the linguistic computation it requires. We use standard terminology from phonology and morphology.

Table 1: The six phonological operations tested, with linguistic classification, example pairs, and dataset sizes.

Op	Linguistic term	Example	Computation	N pairs
1	Rhyming hypocorism	Richard → Dick	Irregular stored lookup	32
2	Clipping (apocope)	Timothy → Tim	Regular first-syllable truncation	90
3	Initialism + letter name	J → Jay	Grapheme → phoneme name	30
4	Oronym (juncture reanalysis)	Bill Ding → building	Cross-boundary phoneme fusion	99
5	Homophone recognition	Neil → kneel	Phoneme-identity detection	120
7	Folk etymology	microwave → Michael Wave	Reverse decomposition	75

2.2 Dataset Construction

Each pair has the format `{good, bad, target_token, foil_token}`, following the BLiMP minimal-pair methodology (Warstadt et al., 2020). The evaluation metric is logit difference: the model should assign higher probability to the target than the foil at the final position.

Tokenization audit. Following Liao and Shi (2026), we audit every pair for BPE tokenization: target and foil must each be a single BPE token. [PENDING: N/total] pairs pass this filter. Multi-token targets are excluded from circuit discovery but reported as a tokenization bottleneck finding.

Procedural expandability. Ops 2, 3, and 5 can be expanded procedurally using CMU Pronouncing Dictionary (Op 2: syllable-based clipping, Op 3: all 26 letters, Op 5: ∼3000 homophone pairs). Ops 1, 4, and 7 are controlled vocabularies — acceptable for psycholinguistic-style studies as long as the full inventory is reported (Warstadt et al., 2020).

3 Behavioral Validation

Before circuit discovery, we test whether GPT-2 Small can perform each operation at all. Gate: >70% accuracy on logit-diff evaluation.

Table 2: Behavioral validation results. Gate threshold: 70% accuracy.

Operation	Accuracy	Mean LD	N	Gate
Op 1: Rhyming hypocorism	[PENDING: xx%]	[PENDING: ±x.xx]	32	[PENDING: PASS/FAIL]
Op 2: Clipping	[PENDING: xx%]	[PENDING: ±x.xx]	90	[PENDING: PASS/FAIL]
Op 3: Letter name	[PENDING: xx%]	[PENDING: ±x.xx]	30	[PENDING: PASS/FAIL]
Op 4: Oronym	[PENDING: xx%]	[PENDING: ±x.xx]	99	[PENDING: PASS/FAIL]
Op 5: Homophone	[PENDING: xx%]	[PENDING: ±x.xx]	120	[PENDING: PASS/FAIL]
Op 7: Folk etymology	[PENDING: xx%]	[PENDING: ±x.xx]	75	[PENDING: PASS/FAIL]

Expected results. Ops 1 and 2 (name → nickname) should pass easily — these are well-known associations in GPT-2's training data. Op 5 (homophones) should pass based on PhonologyBench results. Op 4 (oronym) is uncertain — this requires cross-boundary phoneme fusion, which is bottlenecked by tokenization. Op 7 (folk etymology) is expected to fail on GPT-2 Small since it requires generating creative decompositions not in training data.

Cross-model sweep. For operations that fail the gate on GPT-2 Small, we test GPT-2 Medium and Large. [PENDING: Report which ops pass at which scale.]

4 Circuit Discovery

For each operation passing the behavioral gate, we run three complementary discovery methods:

4.1 EAP (Edge Attribution Patching)

Gradient-based head importance via retained gradients on `hook_z` activations. Ranks all 144 heads by attribution score.

4.2 Activation Patching

Single-head mean ablation: replace each head's activations with corpus mean and measure logit-diff degradation.

4.3 Weight-Space Features

Extract 173 weight features (108 spectral + 65 behavioral proxy) per head. Bootstrap stability classification with greedy formula search identifies the weight-predicted circuit (Tower, 2026).

4.4 Results

[PENDING: Per-operation tables with: (a) EAP top-15 ranking, (b) activation patching top-15, (c) weight-predicted circuit, (d) P/R/F1 of weight prediction vs activation ground truth.]

4.4.1 Op 1: Rhyming Hypocorism Circuit

[PENDING: Expected: lookup circuit. Small number of heads with high OV cosine similarity between name input direction and nickname output direction. Likely concentrated in layers 3–8 (mid-layer, where Liao & Shi find phonological info peaks). Should resemble IOI name-mover heads but for associative content rather than positional copying.]

4.4.2 Op 2: Clipping Circuit

[PENDING: Expected: positional/syllable-boundary detector circuit. Related to acronym circuit's positional heads but operating on within-token phonological structure.]

Table 3: Op 1 circuit: top heads by activation patching.

Head	Patching effect	Weight-predicted?
[PENDING: L?H?]	[PENDING: x.xxx]	[PENDING: Y/N]
[PENDING: L?H?]	[PENDING: x.xxx]	[PENDING: Y/N]
[PENDING: ...]		

4.4.3 Op 3: Initialism + Letter Name Circuit

[PENDING: This is the micro-version of the acronym task. Key test: do the García-Carrasco et al. 8 letter-mover heads appear in this circuit? If yes, convergent validation with independent method. If no, Op 3 requires additional phoneme-lookup heads beyond positional extraction, which would be a new finding.]

4.4.4 Op 4: Oronym Circuit (Core Novel Finding)

[PENDING: This is the most novel circuit. The model must fuse phonemes across a word boundary ("Bill" + "Ding" → "building"). No prior work has identified cross-boundary phoneme fusion heads.

Expected circuit structure:

- Backbone (L0): universal infrastructure
- Mid-layer fusion heads (L3–L8): the novel component — heads that attend across the word boundary and compose phonemic representations
- Readout (L10–L11): standard output infrastructure

Key question: does this circuit overlap with the clipping circuit (Op 2)? If the fusion heads are distinct from clipping heads, that's evidence for specialized phonological circuitry.]

4.4.5 Op 5: Homophone Circuit

[PENDING: Expected: phoneme-identity detection circuit in mid-layers. Liao & Shi's probing results (63–79% rhyme detection accuracy) suggest phonological representations exist in layers 3–8. The homophone circuit should read from these representations.]

4.4.6 Op 7: Folk Etymology Circuit

[PENDING: Expected to fail behavioral gate on GPT-2 Small. If it passes on Medium/Large, the circuit would be the reverse of Op 4 — decomposing a word into name-like constituents rather than composing names into a word.]

5 Cross-Reference Analysis

5.1 Shared Infrastructure

[PENDING: Does the backbone tier (L0H8, L0H9, L0H11) appear in all phonological circuits, consistent with the "operating system" hypothesis from Paper A? Expected: yes, based on transfer results across RTI/SVA/IOI/gendered pronoun.]

5.2 Overlap with Acronym Circuit

The García-Carrasco et al. acronym circuit consists of 8 letter-mover heads. Op 3 (initialism) is the closest match to their task.

Table 4: Overlap between phonological circuits and known circuits.

	RTI	IOI	SVA	Acronym	Induction
Op 1 (hypocorism)	[PENDING: N]	[PENDING: N]	[PENDING: N]	[PENDING: N]	[PENDING: N]
Op 2 (clipping)	[PENDING: N]	[PENDING: N]	[PENDING: N]	[PENDING: N]	[PENDING: N]
Op 3 (letter name)	[PENDING: N]	[PENDING: N]	[PENDING: N]	[PENDING: N]	[PENDING: N]
Op 4 (oronym)	[PENDING: N]	[PENDING: N]	[PENDING: N]	[PENDING: N]	[PENDING: N]
Op 5 (homophone)	[PENDING: N]	[PENDING: N]	[PENDING: N]	[PENDING: N]	[PENDING: N]

5.3 The Phonological Circuit Overlap Matrix

[PENDING: Heat map showing pairwise overlap between all 6 operation circuits. Expected: Ops 1 and 2 (both name-shortening) share more heads than Ops 4 and 5 (different mechanisms). If Op 4 and Op 5 share mid-layer heads, that suggests a shared "phonological representation" layer.]

6 Compositional Validation

[PENDING: Test compound puns that chain operations:

- "Richard Tater" → "Dictator" = Op 1 (Richard → Dick) + Op 4 (Dick Tater → Dictator)
- "Elizabeth Aardvark" → "Lizard" = Op 2 (Elizabeth → Liz) + Op 4 (Liz Ard → Lizard)
- "William Dingo" → "Building" = Op 1 (William → Bill) + Op 4 (Bill Ding → Building)

For each compound: (a) test behavioral accuracy, (b) ablate Op 1 circuit only — compound should break, Op 4 alone should survive, (c) ablate Op 4 circuit only — compound should break, Op 1 alone should survive. This is the compositionality test.

NOTE: This section requires running compound ops after individual op circuits are found. Not included in initial pod run.]

7 Tokenization as Bottleneck

[PENDING: Following Liao & Shi (2026):

- Report STAD scores for all input words
- Test slash-delimited condition ("B/i/l/l D/i/n/g") for Op 4
- If slash-delimiter significantly improves Op 4 accuracy, that's evidence that the oronym circuit *exists* but is bottlenecked by tokenization — the model has the structural capacity (weight-space circuit) but can't activate it with standard tokenization
- This would be a publishable finding connecting the weight-capacity vs. activation-flow distinction from Paper A to a phonological setting

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8 Discussion

Novelty. No prior work has done circuit-level causal analysis on phonological operations. The closest is the acronym circuit (García-Carrasco et al., 2024), which is positional (letter extraction) rather than phonological. Our work connects the behavioral phonological capabilities demonstrated by Suvarna et al. (2024) and Liao and Shi (2026) to specific attention heads for the first time.

The oronym circuit as the core finding. Cross-boundary phoneme fusion — composing two input tokens ("Bill" + "Ding") into a phonologically coherent output ("building") — is the most novel circuit type. If GPT-2 can do this, the model has learned to represent phonological continuity across token boundaries despite training on purely orthographic input.

Implications for the operating system hypothesis. If the backbone tier participates in all phonological circuits, it extends the "operating system" finding from Paper A/D to a new domain. The backbone provides general-purpose attention infrastructure; phonological computation is built on top of it, not in parallel.

Limitations.

- Controlled vocabulary for Ops 1, 4, 7 (not procedurally generated).
- GPT-2 Small may fail behavioral gate on hardest operations.
- English-only; phonological operations differ across languages.
- BPE tokenization is a known confounder — results reflect the interaction of phonological knowledge with tokenization artifacts.

Venue target. COLM, Findings of ACL, or MI workshop. The novel operation types, compositional structure, and first-ever phonological circuit identification position this as a meaningful extension of the MI circuits literature.

References

Tower, E. (2026). Reading circuits from weights. *Manuscript*.

García-Carrasco, J. et al. (2024). How does GPT-2 predict acronyms? *AISTATS 2024*. arXiv:2405.04156.

Suvarna, A., Khandelwal, A., and Peng, N. (2024). PhonologyBench: Evaluating phonological skills of large language models. arXiv:2404.02456.

Liao, C. and Shi, P. (2026). How tokenization limits phonological knowledge representation in language models. *ACL 2026*. arXiv:2604.17105.

Warstadt, A. et al. (2020). BLiMP: The benchmark of linguistic minimal pairs for English. *TACL*.

Wang, K. et al. (2023). Interpretability in the wild. *ICLR 2023*.

Olsson, C. et al. (2022). In-context learning and induction heads. *Transformer Circuits Thread*.

Hanna, M. et al. (2023). How does GPT-2 compute greater-than? *NeurIPS 2023*.